Personalized Medicine Through Personal Data Pods

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Scalable Personalized Medicine?

€ Unfortunately, still quite expensive

But why? Sequencing is getting pretty cheap.

Answer: Data privacy is crucial

Genomic data storage is expensive

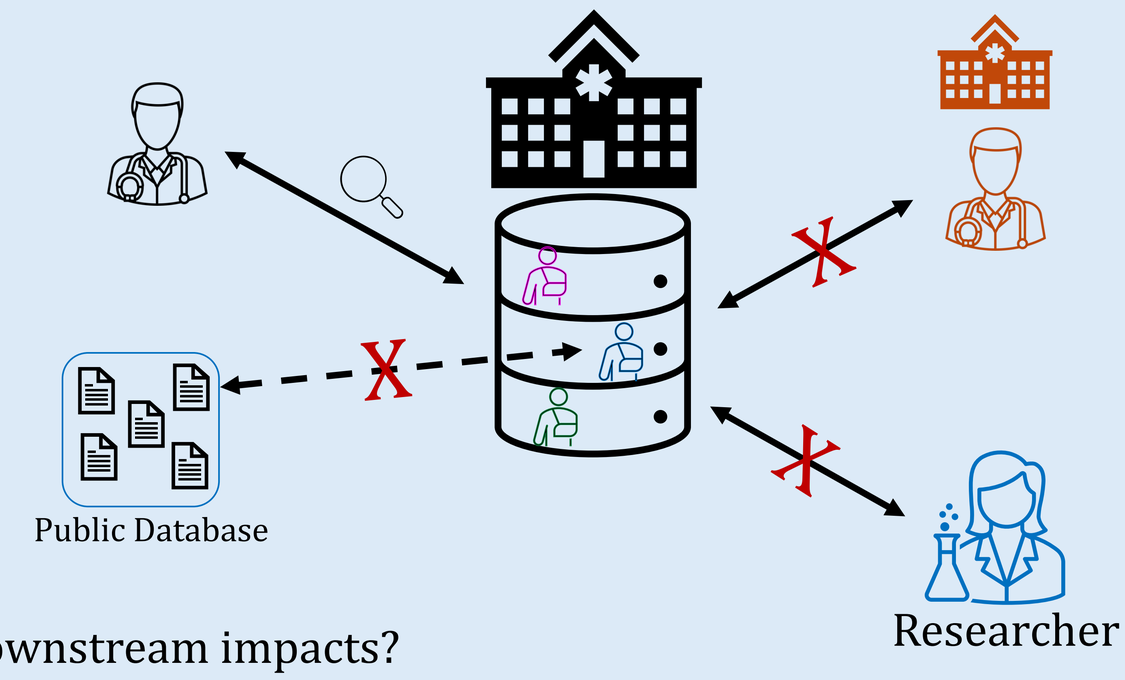
Question: Any way to fix that?

Personal Genome Sequence Data

Very large (~GBs in size)

Require strict privacy protections

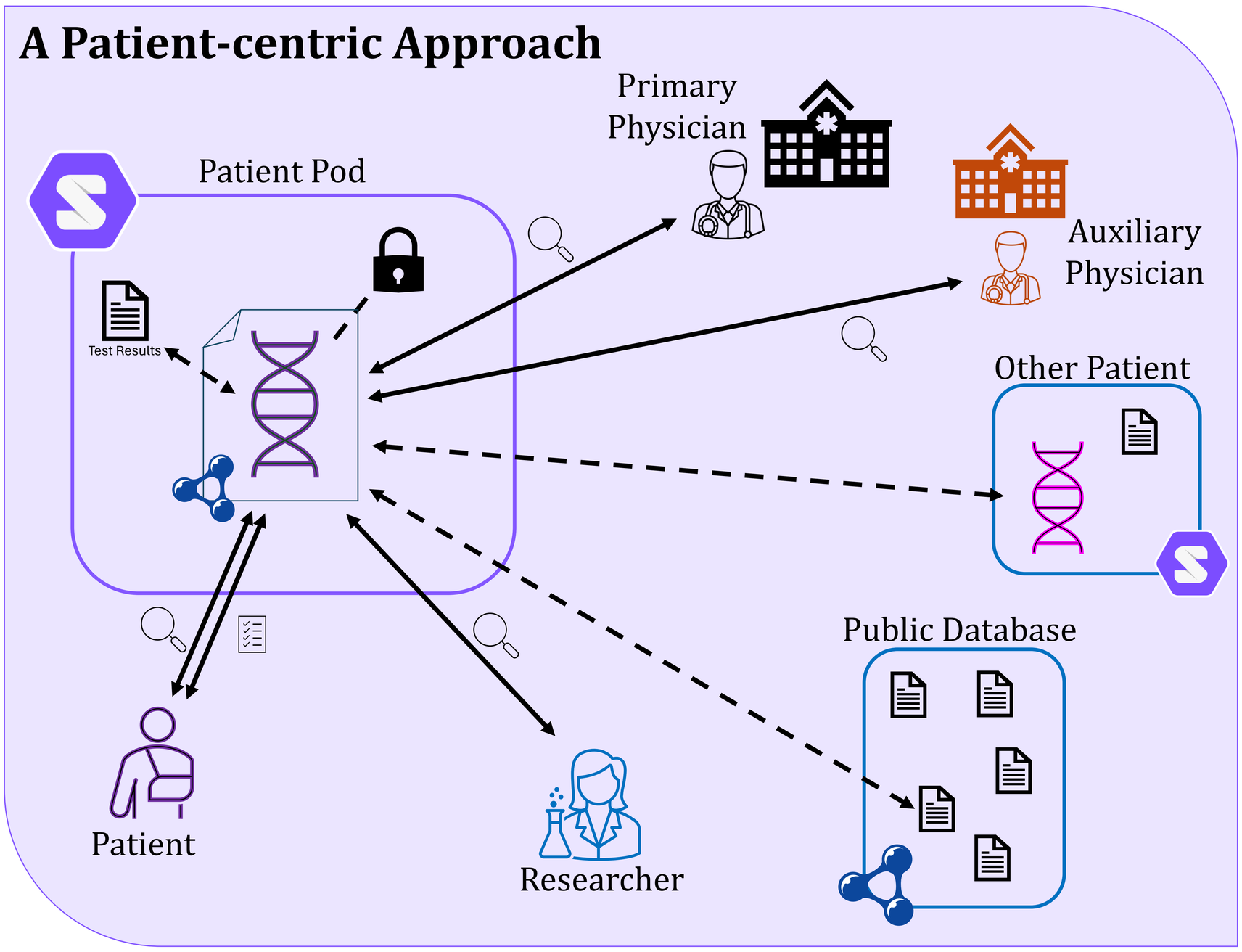
The current institution-centric system



Downstream impacts?

- Reduced data sharing
- Reduced patient transparency
- Increased pertinent data dispersion
- Data duplication

A Patient-centric Approach



Key Challenges

Privacy granularity

- Governance?
- How much patient control?

Data Linkages

- What data should be linked?
- Data formatting Standards?

Querying performance

- Genomic data is BIG
- Requires novel approach?

Poster Paper

Future Directions

Researcher Access

Consent infrastructure

Unified Framework

Contact Details

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